

Poland's Pension Reform: “Security Through Diversity”

*From Multi-Pillar Modernization (1999) to Partial
Reversal (2014)*

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Outline

1. Introduction
2. Overview
3. Rationale for Reform
4. Content of the Reform
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Acronym list

- EC Ageing Report** (European Commission Ageing Report): Long term projections on aging for EU countries
- PAYG DB** (Pay-As-You-Go Defined Benefit)
- NDC**(Notional Defined Contribution): PAYG system with individual notional accounts, contributions are recorded virtually but not invested.
- DC** (Defined Contribution): Pension scheme where benefits depend on contributions and returns; can be funded or notional.
- OFE** (Open Pension Funds) private funded second-pillar pension funds created in 1999.
- ZUS**: Poland's Social Insurance Institution; administers the PAYG/NDC pillar.
- PAYG ZUS**: Poland's Social Insurance Institution; administers the PAYG/NDC pillar.
- FF**: Funded Pillar (Second Pillar) Refers to the mandatory funded component (OFE)

Overview: Socio-economic setting

Rapid population **ageing**:

- The *Ageing Report* shows rising dependency ratios and pension spending stabilising around 10–11% of GDP (long-term).
- Growing share of 65+ in the population.

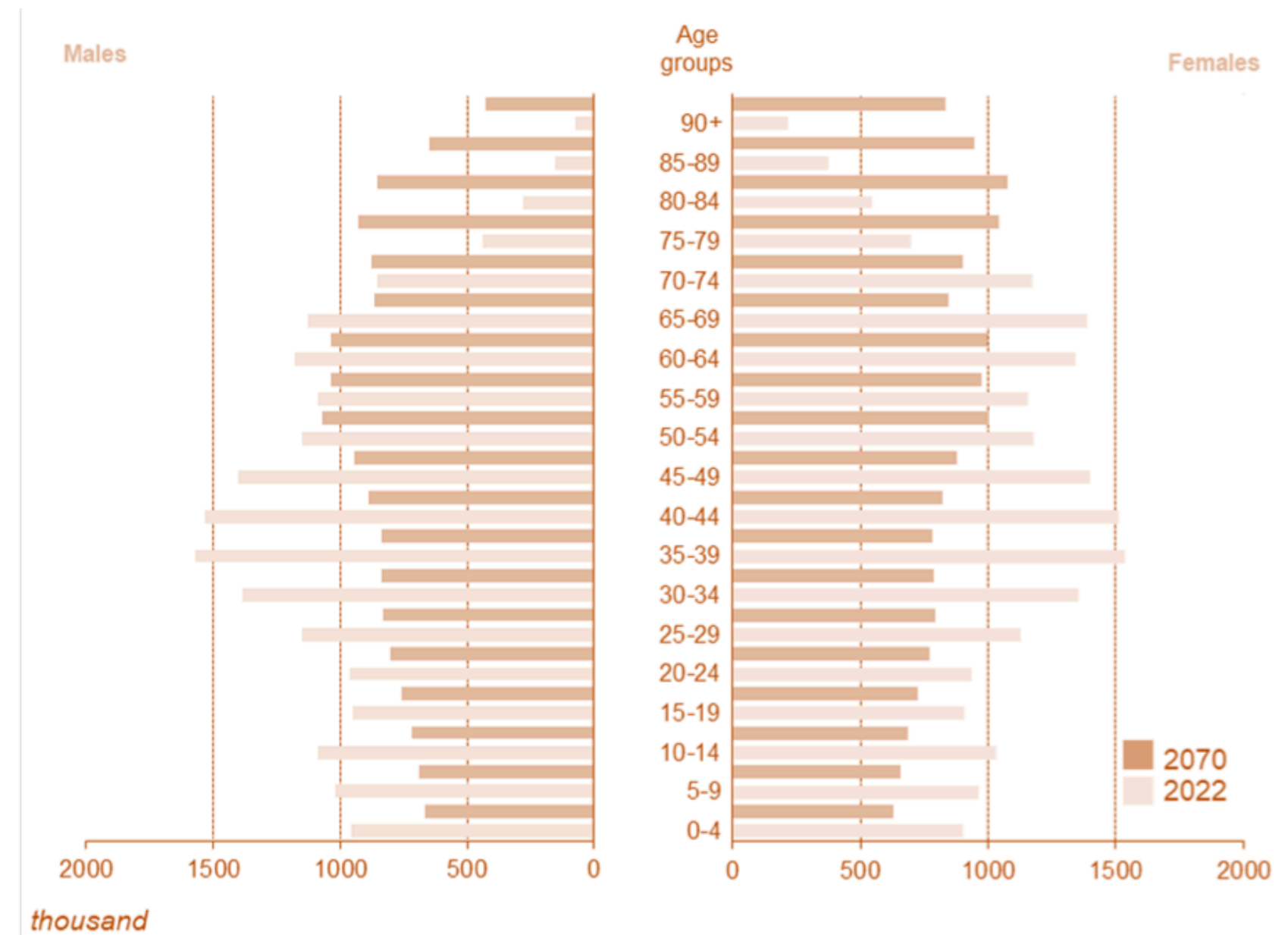


FIGURE 1 – AGE STRUCTURE: 2022 VS 2070

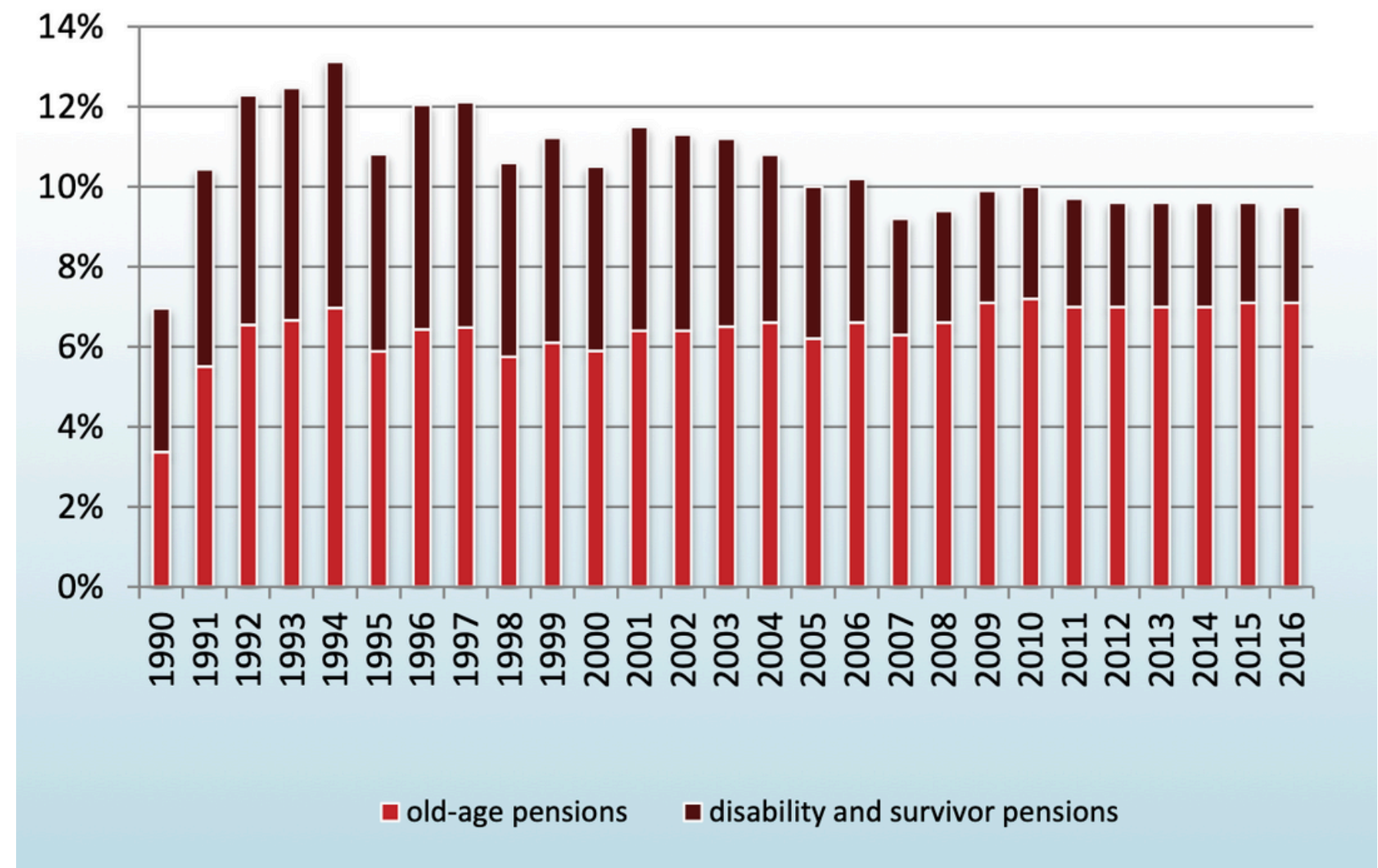
Overview: Socio-economic setting

-High pension spending levels:

- Mid-1990s to early 2000s: pension expenditure $\approx$ 14 to 15% of GDP.
- 2022: about 10.2% of GDP

-Labor market and the early retirement struggle:

- 1990s: low employment of older workers.
- Post-reform improvement:
 - Employment (55–64) rising.



Pension expenditure as % of GDP

Political setting

Post-1989 transition:

- Poland inherited a large, costly PAYG DB system consuming **15% of GDP**.
- Reform concept “**Security through Diversity**” developed with World Bank’s support.
- Broad cross-party political consensus in the late 1990s, enabling the **1999 multi-pillar reform**.

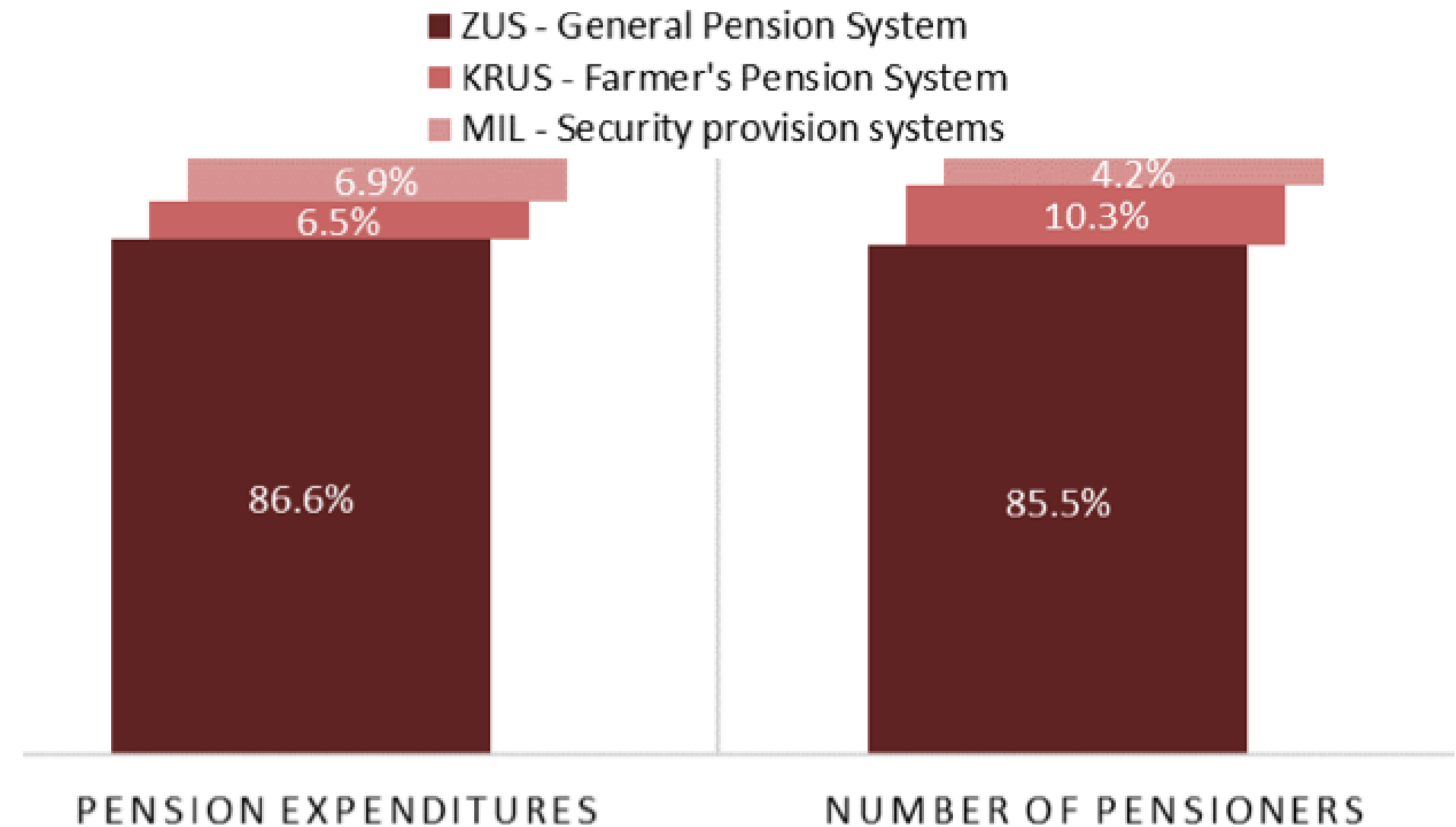


FIGURE 2 –THE STRUCTURE OF THE POLISH PENSION SYSTEM

The Problem (pre-1999)



Costly and **unsustainable** system: with pension *spending* reaching 15% of GDP, **very early** retirement (55/60) and **high** replacement rates

Multiple **early-exit options** and weak incentives to work longer:

- **very low** employment among older workers as a result

PAYG?

“The *pay-as-you-go pension system* is based on the principle of *solidarity*, whereby each **working population** finances the retirement of pensioners by means of a **contribution** that varies according to the economic and demographic situation.”

LECHHAB , L. et DAOUI , D. 2024.

Rationale of the reform

Demographic pressure and aging

- In Poland, as in the rest of Europe, **declining fertility** and **rising life expectancy** made the pay-as-you-go system unsustainable in the long run.
- The retirement of the baby-boom generation further intensified the imbalance.

Inadequacy of ad-hoc adjustments

- **Short-term** measures (indexation changes, benefit recalculations, expansion of early retirement schemes) **were not sufficient** to restore sustainability.
- Structural reforms became unavoidable.



Rationale of the reform

Equity and incentives

- The reforms **strengthened** the link between **contributions** and **benefits**, reducing **implicit redistribution** and **creating incentives** to extend working lives.
- They also aimed to limit early retirement and encourage longer participation in the labor market.

Alignment with international standards

- International institutions (World Bank, OECD, European Commission) encouraged **diversifying** funding sources and introducing more transparent contributory systems.
- The goal was to shift from a single-pillar system (PAYG defined-benefit) to a **multi-pillar system** (NDC, FDC, and voluntary savings).

Content of the 1999 Multi-Pillar Reform

 Replace an *unsustainable* Pay-as-You-Go system with a *diversified*, contribution-based structure.

The new architecture introduced three pillars:

- **Pillar 1:** Notional Defined Contribution PAYG public → administered by ZUS
- **Pillar 2:** Mandatory Financial Defined Contribution → (OFE private funds)
- **Pillar 3:** Voluntary savings → occupational or individual accounts.

Content of the 1999 Multi-Pillar Reform

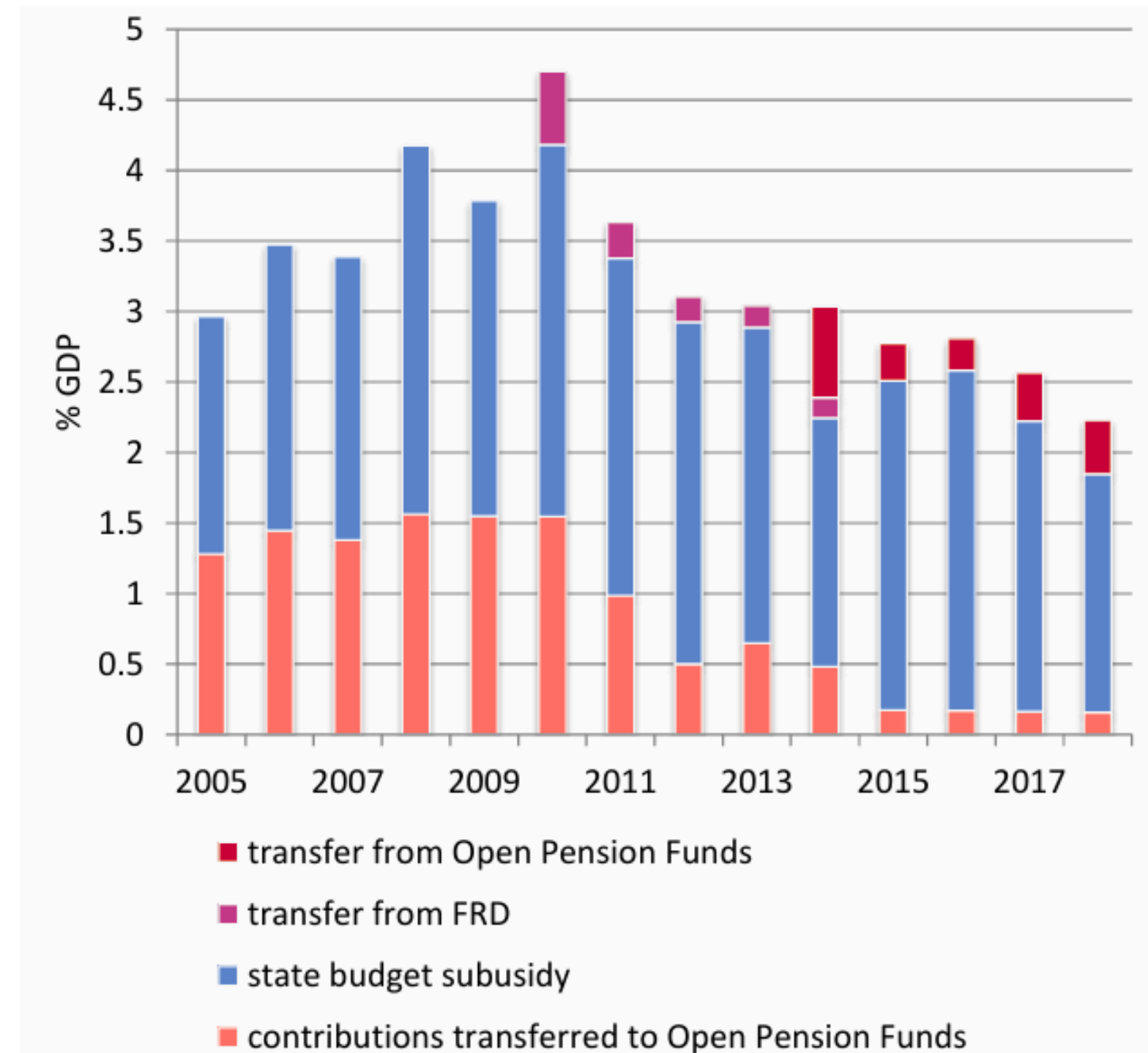
Contribution Split by Pillar

Pillars	Role	Contribution (1999)	Contribution After 2014	Change
Pillar 1 – NDC (ZUS)	Public PAYG notional accounts	12,22%	12,22%	Stable
Pillar 2 – OFE (Funded DC)	Private investment pillar	7.3%	2.92%	Sharp reduction
Pillar 3 – Voluntary	Optional savings	0% mandatory	Outside mandatory	No change

Content of the 1999 Multi-Pillar Reform

Fiscal Impact of **Pillar 2**

- Diverting 7.3% toward OFE created a transition financing gap
- Required state subsidies + borrowing
- After 2014, part of OFE assets reabsorbed by ZUS



Source: Chłoń-Domińczak (2019), Pension Reform – Poland

Implementation & Actors

Key Milestones (1997–1999)

- 1997: Laws establishing OFE and employee pension schemes
- 1998: NDC legislation + social insurance acts
- 1999: Launch of multi-pillar system

Key Actors

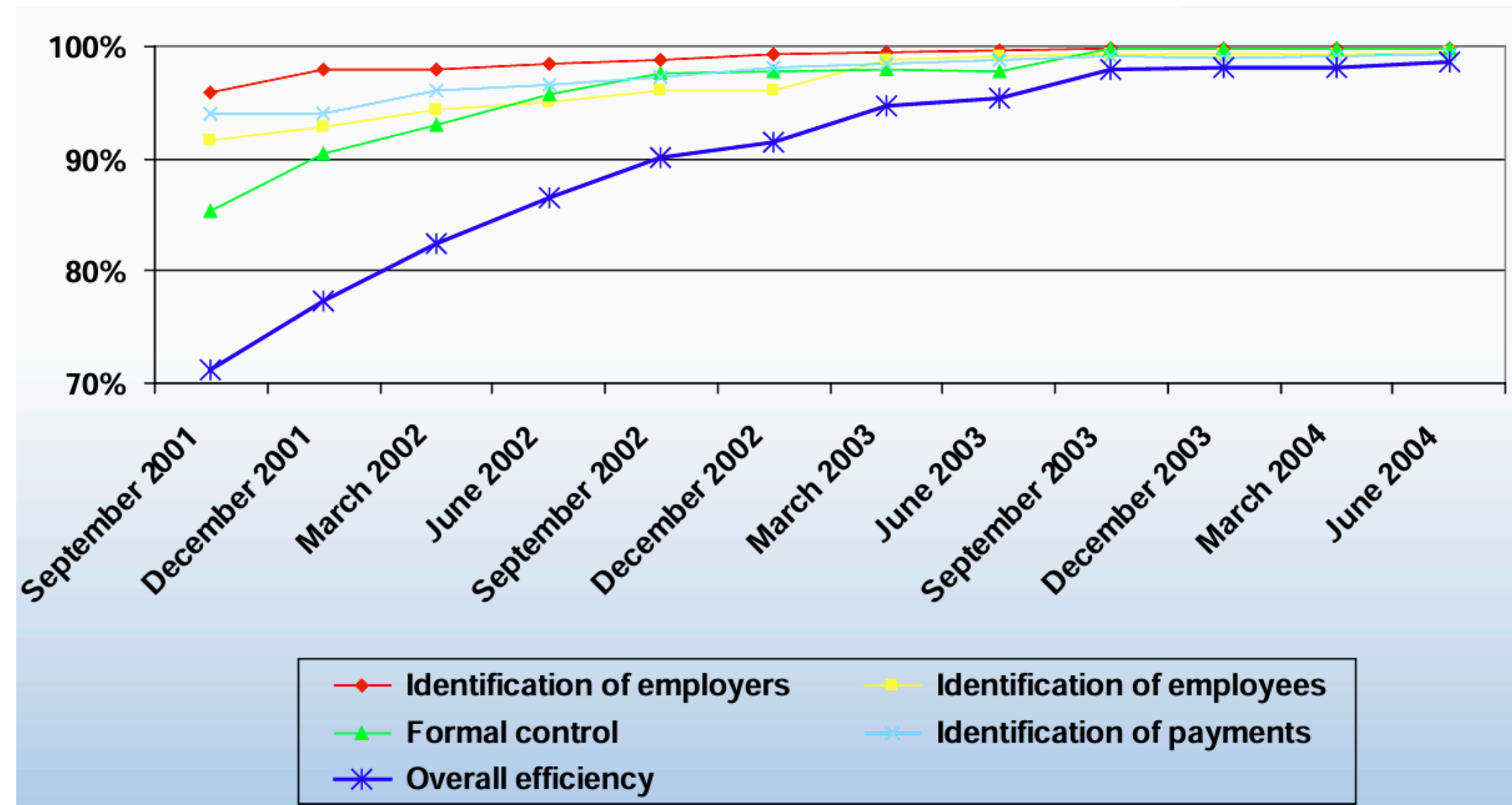
- ZUS: NDC administration & contribution recording
- OFE: Funded DC management
- KNF: Financial supervision
- Employers: reporting contributions
- Government: legislative and fiscal support

Implementation & Actors

Implementation Challenges

- ZUS IT system failures
- Errors in contribution records
- Delays in “initial capital” calculation
- Difficulty matching employer/employee records

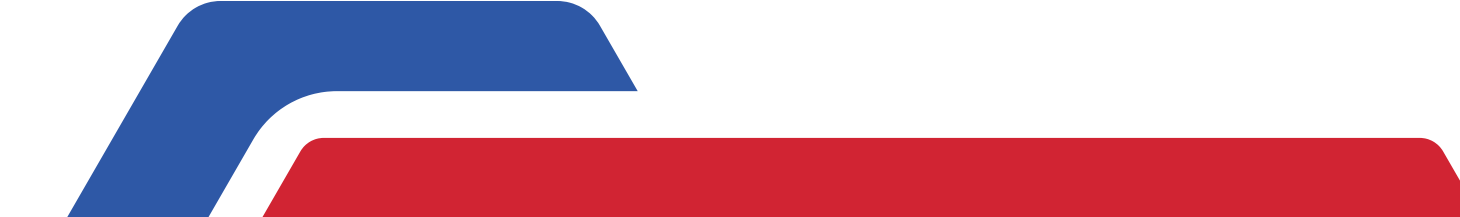
ZUS - Correctness of information



Source: Chłoń-Domińczak (2019), Pension Reform – Poland



The 2014 Reversal?



“The reform that dismantled most of the funded pillar and shifted assets back to the state.”

2014 Reversal: the rationale

Fiscal constraints

- Public debt was approaching the domestic threshold of 55% of GDP.
- OFE held around 145 bn PLN in government bonds ($\approx 8.5\%$ of GDP).

Political drivers

- Post-crisis distrust of financial markets and private pension funds.
- Narrative that OFE charged “high fees” and created “unnecessary fiscal pressure.



Content of the 2014 reversal

Key corrective measures

- Mandatory transfer of 51.5% of OFE assets to ZUS.
- OFE banned from investing in gov bonds.
- Participation becomes voluntary (opt-in) every 4 years.



Assessment



Effectiveness

Efficiency

Equity

Effectiveness

1999

- ✓ Long term sustainability:
 - NDC improves automatic adjustment
- ✓ Diversification:
 - Open pension funds (OFE) created long term institutional investors
- ✓ Labor market effects:
 - Employment of 55–64 increased (World Bank).

2014

Short term gains:

- ✓ • Debt reduction by 8 pp of GDP (IMF 2014).
- ✓ • Nationalisation of 145 bn PLN (8.5% GDP).

Medium/long-term:

- ✓ • Participation in OFE changed from mandatory to voluntary (opt-in).
- ✗ • **Reduced** risk sharing



Assessment

Effectiveness

Efficiency

Equity



Financial Efficiency

Compare the expected returns and variances of the pension funds (OFE) with those of benchmark portfolios.

Result: OFEs have higher returns and lower variances than the benchmarks → evidence of greater efficiency.

■ Table 4. Values of the test-statistics

OFEs	Hypotheses					
	$H_0: E(R_p) = E(R_B)$			$H_0: D^2(R_p) = \sigma_0^2$		
	Portfolio no.			Portfolio no.		
	1	2	3	1	2	3
AEGON	13.582	10.063	5.866	-29.856	-51.174	-46.857
ALLIANZ	14.962	11.434	7.224	-30.013	-51.269	-46.965
AVIVA	13.731	10.503	6.651	-24.98	-48.212	-43.507
AXA	14.595	11.07	6.865	-29.956	-51.235	-46.926
GENERALI	14.654	11.295	7.289	-27.268	-49.602	-45.079
ING	13.822	10.811	7.22	-20.696	-45.61	-40.565
NORDEA	15.118	11.624	7.457	-29.473	-50.941	-46.593
POCZTYLION	12.784	9.516	5.618	-25.692	-48.645	-43.997
PEKAO	13.284	9.97	6.016	-26.514	-49.144	-44.561
PZU	14.072	10.762	6.813	-26.443	-49.101	-44.512
WARTA	13.572	10.38	6.573	-24.292	-47.794	-43.035

Kompany & al, 2015

Financial Efficiency

Sharpe Ratio

It measures the excess return relative to the risk-free rate, scaled by the total volatility of the portfolio.

If Sharpe ratio is positive, the portfolio generates a return above the risk-free rate, adjusted for risk.

■ Table 6. Sharpe ratios calculated for pension funds

Period	AEGON	ALLIANZ	AVIVA	AXA	GENERALI	ING	NORDEA	POCZTYLION	PEKAO	PZU	WARTA
Stagnation 1	0.522	0.531	0.494	0.487	0.476	0.463	0.587	0.413	0.394	0.551	0.434
Bull market 1	0.548	0.562	0.569	0.607	0.607	0.529	0.527	0.579	0.676	0.561	0.608
Bear market	-0.454	-0.460	-0.468	-0.445	-0.457	-0.434	-0.463	-0.461	-0.538	-0.435	-0.497
Bull market 2	0.577	0.599	0.565	0.580	0.559	0.561	0.596	0.571	0.579	0.521	0.571
Stagnation 2	0.230	0.259	0.249	0.282	0.253	0.260	0.298	0.221	0.236	0.236	0.266

SOURCE: OWN ELABORATION.

Kompany & al, 2015

The OFEs generated more return per unit of risk, in terms of total volatility.

Financial Efficiency

Treynor Ratio

It measures the excess return per unit of systematic, market-related risk.

Treynor > 0 → the portfolio adequately **compensates** for market risk.

A higher Treynor ratio → **greater** efficiency in the use of systematic risk.

Compare the Treynor ratios of the OFEs with those of the benchmark portfolios.

The OFEs consistently outperform the benchmarks, with only a few exceptions during bearish market periods.

■ Table 7. Treynor ratios for pension funds for different portfolios representing market benchmarks in the Sharpe models

Period	AEGON	ALLIANZ	AVIVA	AXA	GENERALI	ING	NORDEA	POCZTYLON	PEKAO	PZU	WARTA
Portfolio 1											
Stagnation 1	0.025	0.027	0.023	0.024	0.022	0.023	0.029	0.020	0.021	0.028	0.021
Bull market 1	0.016	0.017	0.017	0.018	0.018	0.016	0.016	0.017	0.020	0.017	0.018
Bear market	-0.022	-0.023	-0.023	-0.022	-0.022	-0.021	-0.023	-0.023	-0.027	-0.022	-0.025
Bull market 2	0.018	0.019	0.018	0.018	0.018	0.018	0.019	0.018	0.018	0.016	0.018
Stagnation 2	0.007	0.008	0.007	0.008	0.007	0.008	0.009	0.006	0.007	0.007	0.008
Portfolio 2											
Stagnation 1	0.038	0.042	0.035	0.037	0.034	0.035	0.045	0.030	0.033	0.043	0.032
Bull market 1	0.026	0.026	0.027	0.028	0.028	0.025	0.024	0.027	0.031	0.026	0.029
Bear market	-0.037	-0.038	-0.038	-0.037	-0.037	-0.036	-0.038	-0.038	-0.045	-0.036	-0.041
Bull market 2	0.030	0.031	0.029	0.030	0.028	0.029	0.031	0.029	0.029	0.027	0.029
Stagnation 2	0.012	0.013	0.012	0.014	0.013	0.013	0.015	0.011	0.012	0.012	0.013
Portfolio 3											
Stagnation 1	0.032	0.036	0.030	0.031	0.029	0.030	0.039	0.026	0.028	0.036	0.028
Bull market 1	0.022	0.023	0.023	0.025	0.025	0.022	0.021	0.024	0.028	0.023	0.025
Bear market	-0.031	-0.032	-0.032	-0.030	-0.031	-0.030	-0.032	-0.031	-0.037	-0.030	-0.034
Bull market 2	0.026	0.028	0.026	0.027	0.025	0.026	0.028	0.026	0.026	0.024	0.026
Stagnation 2	0.010	0.011	0.011	0.012	0.011	0.011	0.013	0.010	0.010	0.010	0.012

Kompany & al, 2015

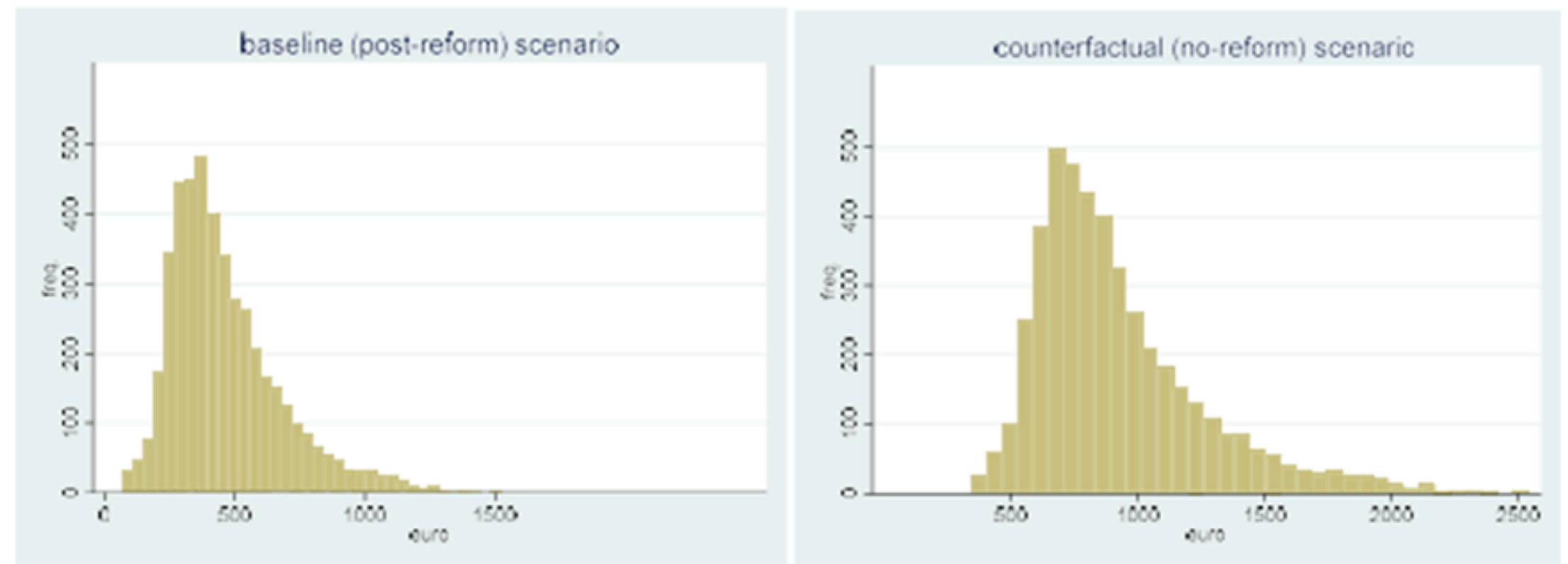
This OFEs generated more return per unit of risk in terms of market risk

Redistributive efficiency

It is weakened as a result. Several empirical studies show that the 1999 reform significantly increases the inequality of future pensions.

For example, the work of Jarocińska and Ruzik-Sierdzińska (2021) indicates that the Gini coefficient of pensions rises from 0.187 in the pre-reform system to about 0.267 for cohorts born in 1979, representing a net increase in inequality generated by the transition toward a more contributory system.

Figure 2. Distributions of predicted monthly pension benefits at 65, baseline scenario (post-reform) and counterfactual (no-reform) scenario, in euro



Note: Exchange rate 2012: 1 EURO = 4.185 PLN.

E.Jarocinska & al, 2021

Macro economic efficiency

Simulations using overlapping generations (OLG) models such as those by Hagemeyer & al (2013) show that the introduction of a funded pillar increases the capital stock and improves potential output in the long term, representing a substantial gain in economic efficiency

However, these gains mainly materialize in the decades following the implementation of the reform and do not fully benefit the generations close to retirement at the time of the change.

At the macroeconomic level, the reform produces positive effects on capital accumulation and growth.



Assessment

Effectiveness

Efficiency

Equity



VERTICAL EQUITY

- Pensions directly **reflect lifetime earnings**.
- **Low-income** and unstable workers accumulate fewer contributions → lower pensions.
- **Limited** redistributive elements in NDC.
- Income gaps in the labor market are reproduced at retirement.
- Overall: The reform **weakens** vertical (redistributive) equity





GENDER EQUITY

- Women accumulate fewer contributions (lower wages, career breaks).
- Longer life expectancy → NDC divides account over more years.
- Funded pillar does not reduce gender disparities.
- Result: structural and persistent gender pension gap.

INTERGENERATIONAL EQUITY

- Cohorts treated differently by design:
 - Before 1949: Protected by generous DB rules
 - 1949–1968: Mixed system
 - After 1968: Fully exposed to NDC + OFE volatility
 - Younger workers take on the fiscal burden of transition.
 - Contribution diversion → PAYG deficit financed by taxpayers (mostly newer generations).
- 

Conclusion

The 1999 reform strengthened the sustainability, effectiveness, and efficiency of the pension system by introducing a defined-contribution structure that improved transparency, actuarial fairness, work incentives, and long-term financial stability. However, it also increased inequality and lowered replacement rates, raising the risk of future old-age poverty.

The 2014 reform focused on short-term fiscal relief by reducing the importance of the funded pillar. Although it temporarily improves public finances, it undermines long-term sustainability, lowers national savings, slows economic growth, and only slightly enhances redistribution, making it ineffective and inefficient over time.

Overall, the 1999 reform is economically efficient but socially costly, whereas the 2014 reform is fiscally driven, only partly effective in the short term, and inefficient in the long term. Future reforms must balance sustainability and fairness through stronger solidarity mechanisms, better pension adequacy, and greater institutional stability.

Sources

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Thank You